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Subject Code	Subject Name	L	T	P	C	QP
23BCSPE47011	Cryptography and Network Security	3	0	0	3	A
UNIT: I		09 Hours				
Introduction: Security Goals, Attacks, Services and Mechanisms, Techniques, CIA Triad, Mathematics of Cryptography: Integer arithmetic, Modular arithmetic, Matrices, Linear Congruence, Algebraic Structure (Groups, Ring, Field), Traditional and Modern Symmetric-Key Ciphers: Substitution, Transposition, Stream and Block Ciphers, Steganography.						
UNIT: II		09 Hours				
DES Structure and Analysis, Multiple DES, Security of DES, AES Structure and Analysis, Transformations, Key expansion, Ciphers, and Use of modern block Ciphers (ECB, CBC, CFB, OFB, CTR Mode).						
UNIT: III		09 Hours				
Euler's Theorem, Fermat's Little Theorem, Primes, Primality Testing, Factorization, Chinese remainder Theorem, Quadratic Congruence, Exponentiation and Logarithm, Asymmetric Key Cryptography: An Overview of Asymmetric Key Cryptography, RSA, Rabin, Elgamal and Elliptic Curve Cryptosystems.						
UNIT: IV		09 Hours				
Message Integrity, Random Oracle Model, Message Authentication, Cryptographic Hash Functions, SHA-512, Whirlpool, Digital Signature: Comparison, Process, Services, Attacks on Digital Signature, Digital Signature Schemes.						
UNIT: V		09 Hours				
Entity Authentication: Password, Challenge Response, Key Management: Symmetric Key Distribution, KERBEROS, Public Key Distribution, Network Security: E-Mail, Firewalls, IDS, PGP, S/MIME, Overview of IP Security, TLS / SSL Architecture, X.509 Certificate.						
Teaching Methods: Chalk & Board / PPT / Video Lectures						
Text Books:						
1. Cryptography & Network Security by Behrouz A. Forouzan, Tata McGraw Hill. 2. Cryptography and Network Security: Principles and Practice by William Stallings, 8th Edition - Pearson						
Reference Books:						
1. Cryptography and Network Security, 4th Edition by Atul Kahate. 2. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition. 3. Cryptography and Network Security: Demystifying the ideas of Network Security, Cryptographic Algorithms, Wireless Security, IP Security, System Security, and Email Security. by Bhushan Trivedi, Savita Gandhi, Dhiren Pandit, BPB Publications.						

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